HD PLATINUM 2.4 GHz RADIO CARRIER

REV A | ENGINEERING FIRST ARTICLE — NOT PRODUCTION RELEASED

CoreS3-SE K128-SE + M032 frame + Base LAN v1.2 K012-B-V12

HOST / STACK-THROUGH BUS

M5-Bus A089

J1

GND	1	GND	G10_/_RF_IRQ	2	RF_IRQ
GND	3	GND	G8_/_RF_MISO	4	MISO_HOST
GND	5	GND	RESET	6	X
	7 X	G37_/_ETH_MOSI	G5_/_RF_MOSI	8	MOSI_HOST
	9 X	G35_/_ETH_MISO	G9_/_ETH_CS	10 X	
	11 X	G36_/_ETH_SCK	HOST_3V3	12	HOST_3V3
	13 X	G44_/_UART0_RX	G43_/_UART0_TX	14 X	
CSN_HOST	15	G18_/_RF_CSN	G17_/_RF_CE	16	CE_HOST
	17 X	G12_/_SYS_SDA	G11_/_SYS_SCL	18 X	
	19 X	G2_/_PA_SDA	G1_/_PA_SCL	20 X	
SCK_HOST	21	G6_/_RF_SCK	G7_/_ETH_RESET	22 X	
	23 X	G13_/_MIC	G0_/_BOOT	24 X	
	25 X	HPWR_/_HOST_NC	G14_/_ETH_IRQ	26 X	
	27 X	HPWR_/_HOST_NC	BUS_5V	28	BUS_5V
	29 X	HPWR_/_HOST_NC	BAT	30 X	

Unused contacts marked X have NO carrier tap.
Their top/bottom stack-through connection remains.
Do not ground HPWR contacts 25/27/29.

Ethernet: MOSI37/MISO35/SCK36/CS9/RST7/IRQ14.
GPIO35 is also LCD D/C: firmware must arbitrate.
Disable SD and audio for initial qualification.

DEDICATED 3.3 V RADIO SUPPLY

FB1 — 600R@100MHz

U2 — TLV75533PDBVR

C1 — 10uF 25V X5R

C2 — 100nF 50V X7R

C3 — 10uF 25V X5R

R9 — 100k 1%

C1/C2 at U2 IN; C3 at U2 OUT.
U2 EN follows host rail; RF_3V3 is NOT tied to HOST_3V3.
All MLCC values are nominal; account for DC-bias derating.

Power flags identify external supply rails / continuity through FB1.

DEDICATED RADIO SPI / DEFAULT STATES

R1–R4 near host connector; R5 near radio.
CE low, CSN high only after RF_3V3 is stable.
Initial SPI: mode 0, 1 MHz. Reserve Ports B/C.

U1 — RF2401F20

R1 33R 1% CE_HOST — RF_CE

R2 33R 1% CSN_HOST — RF_CSN

R3 33R 1% SCK_HOST — RF_SCK

R4 33R 1% MOSI_HOST — RF_MOSI

R5 33R 1% MISO_HOST — RF_MISO

RF_IRQ

R6 — 100k 1% RF_CE

R7 — 100k 1% RF_3V3 / RF_CSN

R8 — 100k 1% RF_SCK

J2 — U.FL — ANT

50 ohm controlled trace.
No RF test-pad stubs.
U.FL to external antenna.
ANT is pin 11, never pin 9.

C4/C5 at U1 VCC/GND

C4 — 47uF 10V X5R

C5 — 100nF 50V X7R

NiceRF current V3.1 numbering
RF_PWR <= 2: default RF_PWR = 0.
Do not substitute RF2401PRO or S124R1.

TEST POINTS — no assembled components

TP1 GND — TP2 BUS5V — TP3 RF3V3 — TP4 CE — TP5 CSN — TP6 IRQ — TP7 HOST3V3

Source: hardware/design/board-spec.json
See electrical-evidence.md and manufacturing acceptance documents.
Current carrier / packet-radio combination requires first-article validation.

Engineering first article — not production released

HD Platinum RF

Sheet: /
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Title: HD Platinum RF carrier

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